STAT 641 **Midterm Exam 1 Solution** **Ghosh**

1. (a) To show $\mathcal{F}$ is a σ-algebra, we only need to verify closure under complements and finite unions. The first one follows from the fact that $\Omega \in \mathcal{F}$ and so $A^c = \Omega \setminus A \in \mathcal{F}$ for every $A \in \mathcal{F}$. Since $\mathcal{F}$ is also a π-system, it contains all finite intersections (and hence by (ii) and De Morgan's law, finite unions). Hence, using the proposition 1.1.1, $\mathcal{F}$ is a σ-algebra.

   (b) It is not a π-system - $\{a,b\} \cap \{a,c,d\} = \{a\} \notin \mathcal{F}$. But it is not a σ-algebra since $\{b\} \cup \{c,d\} = \{b,c,d\} \notin \mathcal{F}$. Also, it can not be a λ-system, since $\{a,b\} \setminus \{b\} = \{a\} \notin \mathcal{F}$.

   (c) Clearly $\mu(A) \in [0,\infty]$ and $\mu(\emptyset) = 0$ are satisfied. Let $\{A_n\} \subset \mathcal{F}$ **and** $\cup A_n \in \mathcal{F}$.
   - if $\cup A_n$ is finite then there has to be finitely many A_is in $\{A_n\}$ that are finite and nonempty, and $\mu(\cup A_n) = \sum \mu(A_n) = |\cup A_n| < \infty$ (this is a finite sum in this case).
   - if $\cup A_n$ is cofinite then $\mu(\cup A_n) = \infty$. If all A_is are finite, then there has to be a countable (infinite) number of these that are nonempty with $\mu(A_i) \geq 1$. Hence, $\mu(\cup A_n) = \sum \mu(A_n) = \infty$ (the sum is an infinite sum with finite summands). If any of A_is is cofinite, then $\mu(A_i) = \infty$ and hence $\mu(\cup A_n) = \sum \mu(A_n) = \infty$.

2. (a) Clearly $\mu : \mathcal{F} \to [0,\infty]$ and $\mu(\emptyset) = \sum 0 = 0$. If disjoint $\{A_n\} \subset \mathcal{F}$, then $\mu(\cup_{i\geq 1} A_i) = \sum_{n\geq 1} n^2 \mu_n(\cup_{i\geq 1} A_i) = \sum_{n\geq 1}\sum_{i\geq 1} n^2 \mu_n(A_i) = \sum_{i\geq 1}\sum_{n\geq 1} n^2 \mu_n(A_i) = \sum_{i\geq 1} \mu(A_i)$, using the fact that each μ_n is a measure (and changing the order of summation is allowed since the summands are non-negative. This completes the proof.

   (b) Taking $A = \Omega \in \mathcal{F}$, we get $B = \Omega \cap B \in \mathcal{F}_B$. If $A = F \cap B \in \mathcal{F}_B$ for some $F \in \mathcal{F}$, then $B \setminus A = F^c \cap B \in \mathcal{F}_B$, since $F^c \in \mathcal{F}$ (note that $B \setminus A$ is the complement of A as a subset of B). If $\{A_i\} \subset \mathcal{F}_B$, then each $A_i = F_i \cap B$, for some $\{F_i\} \subset \mathcal{F}$. In this case, $\cup_{i\geq 1} F_i \in \mathcal{F}$ and hence $\cup_{i\geq 1} A_i = (\cup_{i\geq 1} F_i) \cap B \in \mathcal{F}_B$. This proves that $\mathcal{F}_B$ is a σ-algebra of subsets of B.

   (c) Recall that complement of a closed set is an open set (and vice versa), and σ-algebras contain complements. So $\mathcal{C} \subset \sigma\langle\mathcal{O}\rangle$ and So $\mathcal{O} \subset \sigma\langle\mathcal{C}\rangle$. This completes the proof.

3. (a) Let $\mathcal{C} = \{(a,b] : -\infty \leq a < b < \infty\} \cup \{(a,\infty), a \in \mathbb{R}\}$. It is a semi-algebra. Define $\mu_F(a,b] = F(b+) - F(a+) = F(b) - F(a)$; $\mu_F(a,\infty] = F(\infty-) - F(a+) = 1 - F(a)$ Then this defines a measure on semi-algebra $\mathcal{C}$. By Caratheodory Extension Theorem, there exists an extension of μ_F, called μ_F^* on $(\Omega, \mathcal{M}_{\mu_F^*}, \mu_F^*)$, where $\mathcal{M}_{\mu_F^*}$ = set of all μ_F^*-measurable sets on $\Omega = \{A \in \mathcal{P}(\Omega) : \mu_F^*(E) = \mu_F^*(E \cap A) + \mu_F^*(E \cap A^C), \forall E \in \Omega\}$. $\mathcal{M}_{\mu_F^*}$ is a σ-algebra containing $\mathcal{C} \Rightarrow \mathcal{B}(\mathbb{R}) = \sigma\langle\mathcal{C}\rangle \subset \mathcal{M}_{\mu_F^*}$. μ_F^* restricted to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is an extension of μ_F.

   (b) This is unique by the uniqueness theorem, since $\mu_F^*(\mathbb{R}) = 1$ (finite measure, hence any other measure that takes same value on $\mathcal{C}$ will take same values on $\mathcal{B}(\mathbb{R}) = \sigma\langle\mathcal{C}\rangle$). This is the Exponential (1) probability measure.

   (c) $\mu_F^*(A) = \mu_F^*(1) - \mu_F^*(0) = 1 - e^{-1}$.